

Team Details

Team Name

GadaElectronics

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Year - 2022-2027

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Year - 2022-2027

Problem Statement

Selected Problem

Real-Time Multi-User Smart Assistant for Dynamic and Noisy Smart Environments

Our Problem Understanding (Use Cases Addressed)

1. The system must **separate overlapping speech from a single input** and identify who is speaking. It should handle overlapping voices, map them to user profiles, and filter out disturbances.
2. Recognize and accurately **detect multiple keywords** spoken simultaneously within overlapping audio streams.
3. After separating voices, the **system must decide: Is this a command, conversation, noise?**. This is where most assistants fail (**false triggers and wrong actions**).
4. When multiple valid commands appear, the system should **resolve conflicts intelligently** and know when to ask for clarification.
5. Personalization must be accurate and context-aware, while still keeping **resource usage efficient**.
6. Existing assistants treat every audio event with **equal computational weight**. **The real challenge** is knowing which audio events **deserve computation at all and escalating resources** only when the scene genuinely demands it.

Proposed Solution

Our Solution – The Canary

“Because not every sound deserves a million parameter computation. Every sound must first earn its compute.”

The Passive Idle Gate

Use Cases Addressed 6, 1

A lightweight layer guards the entire pipeline. It outputs just 3 signals: **speech_detected** | **wakeword_probability** | **noise_floor**. If nothing clears the system stays idle. Supports **personalized wakewords**, **speech deformities**.

Shown in Stage - 0

Confidence-Based Fallback Mechanism

Use Cases Addressed 4

Every module outputs: **predicted intent**, **confidence score**, **signal quality**, and **identity confidence**.

Decides to execute, ignore, or clarify. **It never blindly trusts its own output**.

Shown in Stage - 2, Context Engine

Dynamic Routing

Use Cases Addressed 3, 5

Every acoustic event receives a **Scene Complexity Score (SCS)** based on **probability**, **overlap severity**, **wake-word certainty**, and **speaker count**.

Only high-scoring events escalate.

Shown in: Dynamic Resource Scaler

Multi-Speaker Separation

Use Cases Addressed 2, 3

A **Count-Estimator** predicts active speakers and dynamically routes audio through **Libri2Mix/Libri3Mix** separation.

ECAPA-based identification using embeddings and **acoustic voice signatures**.

Shown in Stage - 1, 2

User Arbitration & Priority

Use Cases Addressed 2, 6

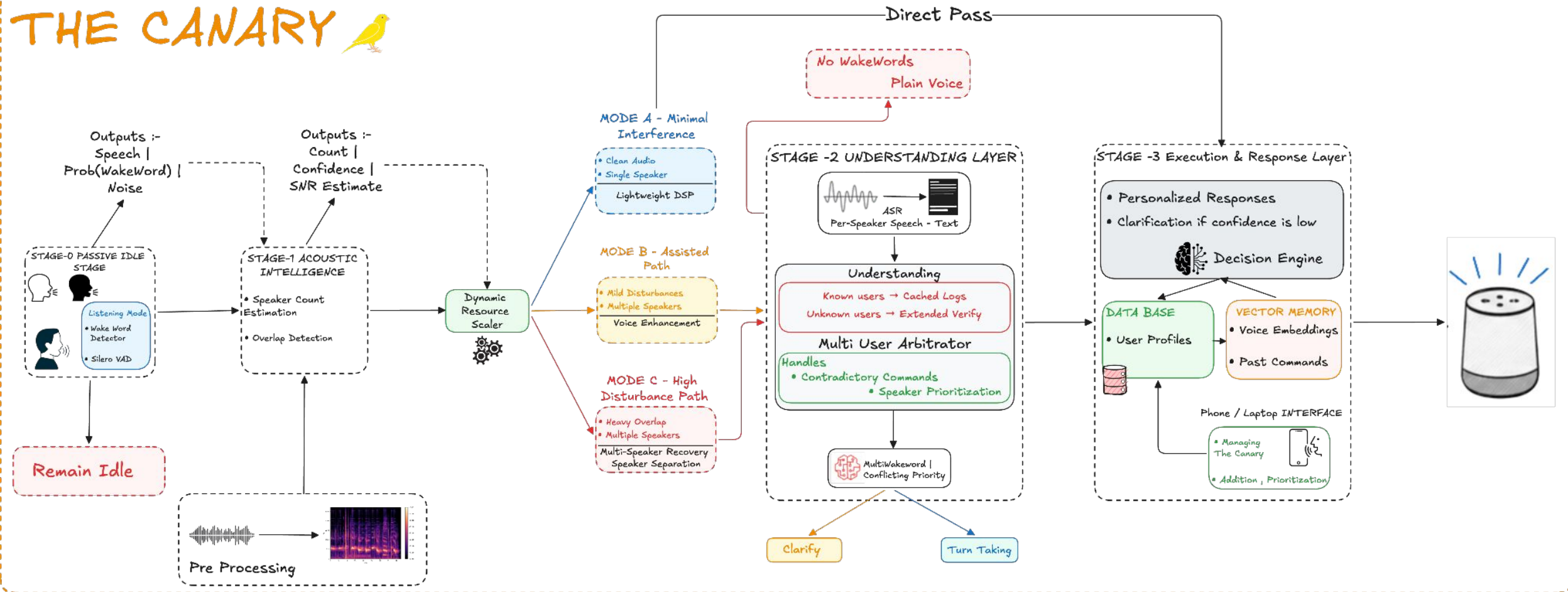
- **Wakeword & Identity Confidence**
- **Speaker Authority**
- **Separation Quality**

Resolves conflicts and enables personalized execution via an **arbitration engine**.

Shown in Stage - 3

Proposed Solution – Overview Of The Entire Technical Architecture

THE CANARY



The Canary first determines **whether an audio event is important enough** to process, using its Passive Idle Gate. It then analyzes the scene through a **Dynamic Resource Scaler (DRS)**, which computes a **Scene Complexity Score** and activates separation only when required. The separated voices are matched against registered profiles using **embeddings**, allowing the system to identify. An **arbitration layer** then resolves conflicts between multiple users and prioritizes commands based on identity & wake-word certainty. Using **speaker identity, confidence scores, & stored preferences**, **Canary** delivers the most relevant response to the right person.

Proof of Working: Scenario Wise Execution

*Terminal screenshot

Scenario A - Clean Single Speaker

DRS ANALYSIS	
Noise Score	: 0.169
Overlap Score	: 0.000
Speaker Score	: 0.000
SCS	: 0.051
Mode	: A 

INFERENCES

Single known user
Low noise
→ Direct execution

Route :  EXECUTE

Scenario B - Multiple Speakers

DRS ANALYSIS	
Noise Score	: 0.281
Overlap Score	: 0.244
Speaker Score	: 0.500
SCS	: 0.306
Mode	: B 

INFERENCES

Two identified users
Multiple valid commands
→ Comparing Intent/Time

Route :  SEQUENTIAL

speaker_1.wav → Hemang Seth
profile: [embe:Hemang pitc:Deep]
speaker_2.wav → Sanchit
profile: [embe:Sanchit pitc:s]

Scenario C - Heavy Noise

x speaker_1.wav [RMS:-18dBFS | Speech:0%] → REJECTED
► speaker_2.wav [RMS:-24dBFS | Speech:46%] → READY -

DRS ANALYSIS	
Noise Score	: 0.988
Overlap Score	: 0.000
Speaker Score	: 0.500
SCS	: 0.396
Mode	: C 

INFERENCES

Heavy noise detected
→ Noise recovery path activated

Speaker : Hemang Seth ◀ WINNER
Known User : YES Wakeword: 1.00

Scenario D - Heavy Noise & Multiple Speakers

DRS ANALYSIS	
Noise Score	: 0.998
Overlap Score	: 0.300
Speaker Score	: 0.500
SCS	: 0.396
Mode	: C 

INFERENCES

Noise + multiple users
→ Arbitration engine, Noise
recovery path resolves priority

Decision :  SEQUENTIAL EXECUTION

What Makes The Canary Different?

"Voice assistants understand commands. Canary understands who said them."



Multi-User Arbitration Engine

Priority-Weighted Execution

Conflicting commands are resolved using identity, wakeword confidence, and household priority.

Conflict Intelligence

Opposing simultaneous **commands trigger clarification, not silent execution**. No command is blindly dropped.

Speaker	: Hemang Seth	WINNER
Known User	: YES	Wakeword: 1.00 Sep.Quality: 1.00
Identity Conf:	0.73	
Intent	: WEATHER	
Priority	: 0.89	$= 0.40 \times \text{wakeword}(1.00) + 0.40 \times \text{id_conf}(0.73)$
Speaker	: Sanchit	
Known User	: YES	Wakeword: 1.00 Sep.Quality: 1.00
Identity Conf:	0.56	
Intent	: WEATHER	
Priority	: 0.82	$= 0.40 \times \text{wakeword}(1.00) + 0.40 \times \text{id_conf}(0.56)$



Speech Accessibility

Adaptive WakeWord Recognition

Maps slurred or mispronounced wake words (e.g., "Kannary $\longleftrightarrow$ Kennedy", "Alexa $\longleftrightarrow$ Aleca") to high confidence scores reliably.

Built for Every Voice / Wakeword

Users can **create personalized wake words**, which are detected using a weighted phonetic matching algorithm that adapts to pronunciation differences and speaking styles.

```
"lookup_table": {  
  "canary": 1.0000,  
  "kanary": 0.9917,  
  "cannary": 0.9857,  
  "cenary": 0.9833,  
  "qanary": 0.9833,  
  "canari": 0.9833,  
  "kannary": 0.9786,  
}
```



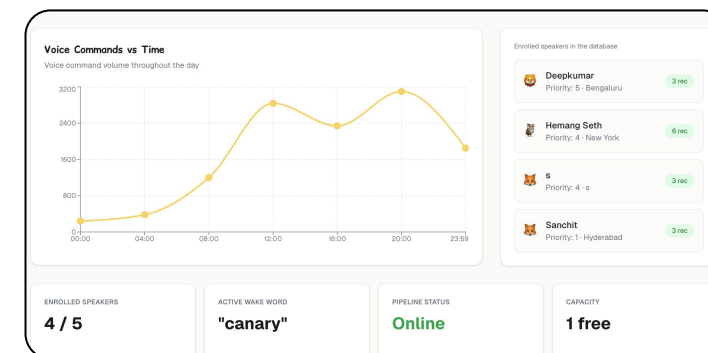
Family Priority Control

Instant Synchronization

A full **web and mobile interface to enroll speakers and set priorities**.

Household Control

Manage the system from any device. Update wake words or family member priorities instantly.



Comparison against Existing Agents

Feature	Alexa	Google Assistant	Siri	Canary
Multi-Speaker Identification	Partial	Partial	Partial	✓
Conflict Resolution	✗	✗	✗	✓
Priority-Based Execution	✗	✗	✗	✓
Personalized Wake Words	Limited	✗	Limited	✓
Accessibility-Aware Wake Words	✗	✗	Partial	✓
Dynamic Resource Scaling	✗	✗	✗	✓
Local User Profiles	Partial	Partial	Partial	✓

Engineering Evaluation & Final Performance

Model Comparison Table

Model	Parameters	xRT Score	Status
SepFormer	25.7M	0.617	- Too Large
DPRNN	3.65M	0.85	- Too Slow
TIGER	0.82M	8.03	- Too Slow
ConvTasNet	5.07M	0.11	✓ Selected

ConvTasNet was the only model capable of real-time CPU inference while maintaining strong separation quality. It also reduced the WER from **70.8% to 23.2%**, which improved command understanding. (xRT of ConvTasNet)

WER Improvement

70.8% → 23.2%

↓ 47.6% Reduction

Reduced transcription errors, enabling reliable command understanding.

Real-Time Performance

xRT = 0.505 (Overall)

~9× Faster

Processes audio faster than playback, helping in better CPU deployment.

Separation Quality

SI-SNR = 14.97 dB

Best quality-to-latency tradeoff among all evaluated models.

Noise Robustness

SI-SNRi = 12.75 dB

Maintains strong separation gains even under noisy conditions.

Evaluation Setup & Methodology

Dataset MiniLibriMix	Test Samples 50 Real 2-Speaker Mixtures	Processing Unit CPU-only Inference	Metric Standard Real-Time Benchmark
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We tested multiple models on **MiniLibriMix** mixtures under our **CPU constraints**.

None simultaneously met the strict parameter, latency, and SI-SNR targets.

KPI Feasibility & Engineering Tradeoffs

"The challenge was not maximizing a single metric, but achieving the best balance between accuracy, latency, model size, and compute efficiency."

Why We Didn't Chase 25 dB

Model	Params	SI-SNR	xRT
ConvTasNet	~5M	15 dB	0.11
DPRNN	3.6M	18–20 dB	0.85
SepFormer	26M	22–24 dB	0.62

Why ConvTasNet?

Real-Time: 0.505 xRT on CPU
Budget: ~5M Params (5.06M)
Quality: 14.97 dB Separation
Stability: Fully Convolutional

The only architecture that satisfied real-time constraints while maintaining separation quality.

Engineering Outcome

Benchmark KPI Target:
25 dB SI-SNR

"An assistant is ultimately evaluated by whether the right user receives the right response at the right time, not by separation quality alone."

Metric	Final Achieved Result
Real-Time CPU Usage	0.505 xRT
Speaker Separation (SI-SNR)	14.97 dB
WER Reduction	70.8% → 23.2%
Multi-User Personalization	Fully Implemented

Optimization & Quality Recovery

Optimization Pipeline

Raw Model 16.3 dB

Chunked Wrapper 11.5 dB (-4.8)

STFT Refine 9.2 dB (-2.3)

Scale-Aware MC 12.8 dB (+3.6)

Gram-Schmidt 16.2 dB (+3.4)

Final Pipeline Output

14.97 dB

Memory-Efficient Processing

Challenge: Audio chunking introduced boundary artifacts. (-4.8 dB SI-SNR)

Additional Speech Refinement

Challenge: Process removed useful speech along with noise. (-2.3 dB SI-SNR)

Signal Reconstruction Improvements

Impact: Preserved speaker info & reduced interference. (+3.6 dB SI-SNR)

Speaker Bleed Reduction

Impact: Cleaner separation during overlaps. (+3.4 dB SI-SNR)

Final Outcome

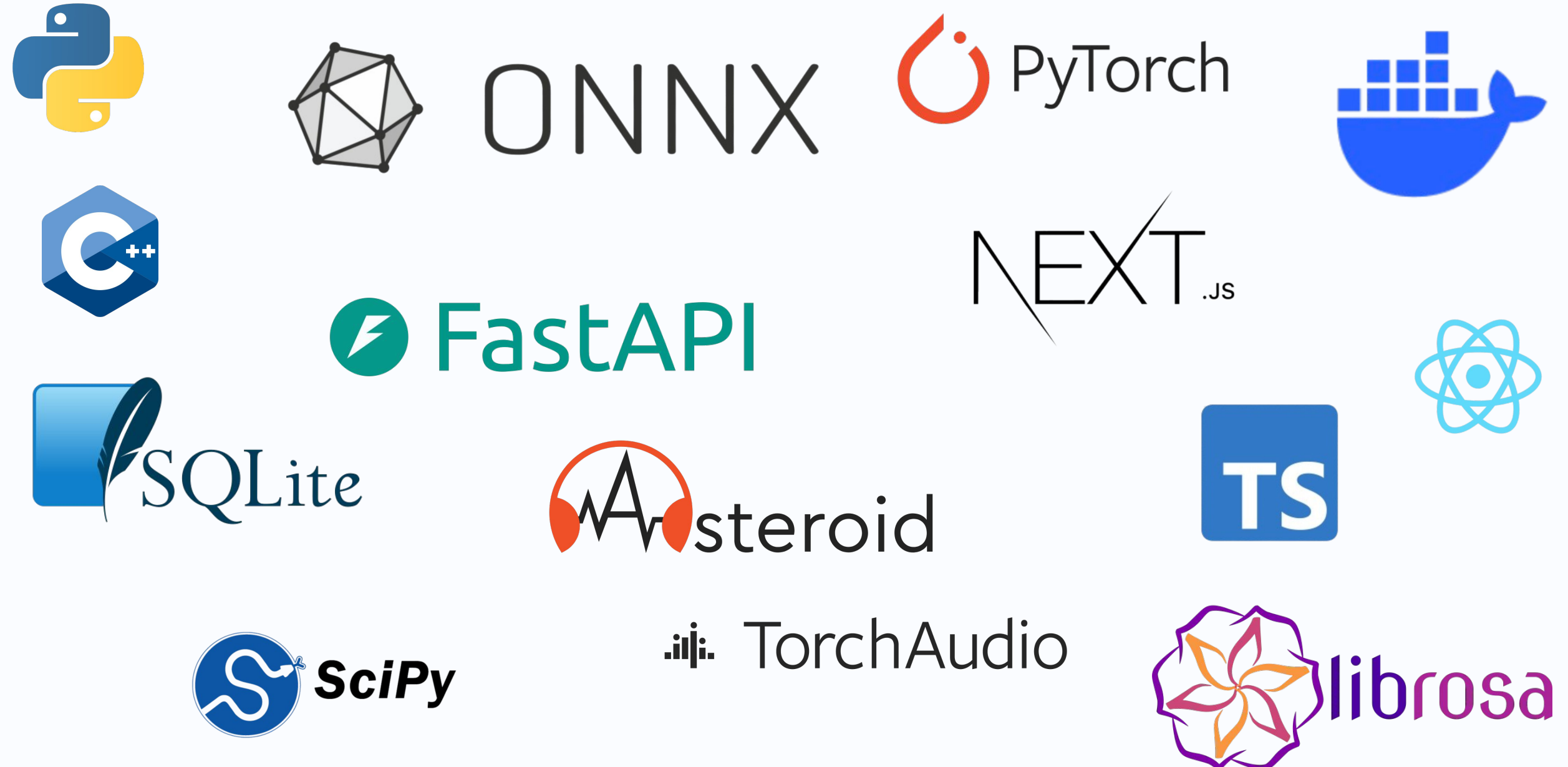
The biggest gains came from improving the processing pipeline rather than changing the neural network itself.

SI-SNR: 11.31 dB → 14.97 dB

SI-SNRi: 8.00 dB → 12.75 dB

xRT: 0.505 (real-time capable)

Tech Stack (Frameworks & Libraries) Used



Open Models / Algorithms / Datasets used

Model	Role	Params	Pipeline Stage
Layer 0: Passive Idle State			
Silero VAD	Voice activity detection + energy floor gating.	0.463M	Continuous listen → Silent (stop) or Activity (→ Layer 1)
Custom C++ Wakeword	Weighted-Levenshtein phonetic wake-word detection algorithm with fuzzy mis-transcription table.	0	Wake confirmed → route to pipeline · not-wake → stop
Layer 1: Acoustic scene intelligence			
Noisereduce	Non-stationary spectral gating denoiser. Removes noise while preserving speech.	0	Pre-separation denoising pass
Spectral Speaker Counter	Greedy agglomerative clustering on 6-dim acoustic features over 500 ms sliding windows. Estimates 1–3 speakers.	0	1 speaker → direct enhance · ≥ 2 speakers → separation
Layer 2: Speaker Separation			
ConvTasNet	Fully convolutional TCN encoder/separator/decoder	~5.006 M	Warm xRT ~0.505 · SI-SNR 14.97 dB clean · SI-SNRi 12.75 dB noisy
Layer 3: Execution and Response Layer			
Whisper tiny (OpenAI)	Automatic speech recognition with pre-screen (energy/VAD) + post-screen (logprob/repetition gates).	37.185M	Audio → Text & Transcript → intent engine

Current Limitations & Future Work

Current Limitations

- 1 Multi-Speaker Scalability**
Optimized for 2-speaker overlap; **performance may degrade in dense 3+ speaker** conversations.
- 2 Cold-Start Inference Latency**
First request after startup **incurs latency for model loading**; subsequent requests at xRT = 0.505.
- 3 Limited Speaker Enrollment Data**
Currently only three short voice samples; **more recordings needed** for robustness across styles.
- 4 Evaluation Scope**
Validated on MiniLibriMix; real-world household recordings and **diverse acoustics** remain future work.
- 5 Separation-ASR Error Propagation**
Residual artifacts can affect downstream ASR in highly overlapping or noisy conditions.

Future Work

-  **Robust 3+ Speaker Support**
Extend **separation and identification** capabilities for dense audio environments.
-  **Adaptive Enrollment**
Introduce **continuous speaker profile refinement** for improved long-term accuracy.
-  **Latency Optimization**
Model preloading and persistent workers to eliminate cold-start delays.
-  **Real-World Evaluation**
Benchmark against larger, more diverse conversational datasets.
-  **Compact Models**
Explore **next-gen separators** satisfying the strict <5M parameter target.

Supporting Research Materials

1. Silero VAD

Lightweight VAD for speech detection and passive pipeline gating.

github.com/snakers4/silero-vad

Used in: Layer 0 – Gating & Adaptive VAD

2. Conv-TasNet

Fully convolutional time-domain audio separation network.

arxiv.org/abs/1809.07454

Used in: Layer 2 – Multi-Speaker Separation

3. Asteroid

Open-source separation toolkit for ConvTasNet and evaluation.

arxiv.org/abs/2005.04132

Used in: Layer 2 – Separation toolkit & model benchmarking

4. LibriMix & MiniLibrimix

Dataset for training and benchmarking speech separation models.

arxiv.org/abs/2005.11262

Used in: Evaluation – SI-SNR & WER

5. ECAPA-TDNN

Speaker embedding model using channel attention.

arxiv.org/abs/2005.07143

Used in: Stage 2 - Speaker Intelligence & Multi-User Arbitration

6. FastDTW

Core of the Acoustic RAG ASR-bypass for dysarthric speech.

arxiv.org/abs/2102.10219

Used in: Layer 4 – Acoustic RAG accessibility fallback

How To Access Our Canary

GitHub Repository

Access the source code, documentation, and development history of The Canary project.

github.com/meikenofdarth/The-Canary

Docker Hub

Pull the latest containerized image for rapid deployment and testing in your environment.

<https://hub.docker.com/r/knightstriker/the-canary>

For technical support or implementation queries, please refer to the README in our GitHub repository.
You can also contact us at Hemang.Seth@iiitb.ac.in / Sanchit.Dogra@iiitb.ac.in